

Continuous Time Heterogeneous Agent Models & Deep Learning

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2026 Turin Summer School on AI for Economics and Finance

August 25, 2026

Outline

1. Huggett (1993) in continuous time.

*Textbook heterogeneous agent model **without** aggregate uncertainty*

2. Krusell and Smith (1998) in continuous time.

*Textbook heterogeneous agent model **with** aggregate uncertainty*

3. Deep learning for continuous time heterogeneous agent model **with** aggregate uncertainty (master equations)

*Reference: Achdou, Han, Lasry, Lion, and Moll (2022) “HACT”;
Payne, Rebei, and Yang (2024) “DeepSAM”*

Huggett (1993)

Definition. The problem of household $i \in [0, 1]$ (in sequence form) is

$$\max_{\{c_{i,t}\}_{t \geq 0}} \mathbb{E}_0 \int_0^\infty e^{-\rho t} u(c_{i,t}) dt \quad \text{s.t.}$$

$$\dot{a}_{i,t} = w_t y_{i,t} + r_t a_{i,t} - c_{i,t}$$

$$y_{i,t} \in \{y_1, y_2\} \text{ Poisson with intensities } \lambda_1, \lambda_2$$

$$a_{i,t} \geq \underline{a}$$

taking as given initial $(a_{i,0}, y_{i,0})$

A **solution** to the household problem is a stochastic process $\{c_{i,t}, a_{i,t}\}_{t \geq 0}$

- A representative firm produces the (homogeneous) final consumption good using technology

$$Y_t = A_t \ell_t$$

- Firms are small and perfectly competitive $\implies$ firm problem is

$$\max_{\ell_t} Y_t - w_t \ell_t$$

- Firm problem is **static** $\implies$ otherwise we would have to think hard about ownership
- Market clearing: $Y_t = \int_0^1 c_{i,t} di, \ell_t = \int_0^1 y_{i,t} di, 0 = \int_0^1 a_{i,t} di$

Competitive Equilibrium

Definition. (Competitive equilibrium: sequence form) *Taking as given an initial distribution of assets and individual labor productivities $\{a_{i,0}, y_{i,0}\}_i$ as well as an exogenous path for TFP $\{A_t\}$, a competitive equilibrium comprises an allocation $\{Y_t, \ell_t, c_{i,t}, a_{i,t}\}$ and prices $\{r_t, w_t\}$ such that: (i) households optimize, (ii) firms optimize, and (iii) markets clear.*

Recursive representation

$$\rho V_t(a, y) = \max_c \left\{ u(c) + \mathbb{E}_t \frac{dV_t(a, y)}{dt} \right\}$$

Q1: What is continuation value?

Recursive representation

$$\rho V_t(a, y) = \max_c \left\{ u(c) + \mathbb{E}_t \frac{dV_t(a, y)}{dt} \right\}$$

Q1: What is continuation value?

$$\rho V_t(a, y_j) = \max_c \left\{ u(c) + (r_t a + w_t y_j - c) \partial_a V_t(a, y_j) + \lambda_j (V_t(a, y_{-j}) - V_t(a, y_j)) + \partial_t V_t(a, y_j) \right\}$$

Resolving max operator gives FOC, which defines **consumption policy function**:

$$u'(c_t(a, y_j)) = \partial_a V_t(a, y_j)$$

for all a and j . Define **savings policy function** as $s_t(a, y_j) = r_t a + w_t y_j - c_t(a, y_j)$

Q2: Where is the borrowing constraint $a_{i,t} \geq \underline{a}$ in the HJB?

Answer: in the boundary condition!

- Borrowing constraint gives rise to **state constraint boundary condition**

$$\partial_a V_t(\underline{a}, y_j) \geq u'(r_t \underline{a} + w_t y_j)$$

- Economic intuition: value of saving must be weakly larger than value of consuming
- Heuristic derivation: the FOC still holds at the borrowing constraint

$$u'(c_t(\underline{a}, j)) = \partial_a V_t(\underline{a}, y_j)$$

- But borrowing constraint requires that

$$s_t(\underline{a}, y_j) = r_t \underline{a} + w_t y_j - c_t(\underline{a}, y_j) \geq 0$$

- Borrowing constraint as boundary condition: one advantage of continuous time

Summary: A solution to the household problem in **recursive form** is a set of two functions $V_t(a, y)$ and $c_t(a, y)$ that satisfy

$$\rho V_t(a, y_j) = u(c_t(a, y_j)) + (r_t a + w_t y_j - c_t(a, y_j)) \partial_a V_t(a, y_j) + \lambda_j (V_t(a, y_{-j}) - V_t(a, y_j)) + \partial_t V_t(a, y_j)$$

$$u'(c_t(a, y_j)) = \partial_a V_t(a, y_j)$$

with HJB boundary condition

$$\partial_a V_t(\underline{a}, y_j) \geq u'(r_t \underline{a} + w_t y_j)$$

To save space, will now use savings policy function $s_t(a, y_j) \equiv r_t a + w_t y_j - c_t(a, y_j)$ as shorthand

Kolmogorov Forward Equation

Result: the joint density $g_t(a, y)$ solves the Kolmogorov forward (KF) equation

$$\partial_t g_t(a, y_j) = -\partial_a \left[(r_t a + w_t y_j - c_t(a, y_j)) g_t(a, y_j) \right] - \lambda_j g_t(a, y_j) + \lambda_{-j} g_t(a, y_{-j})$$

Proof: See note on GitHub.

Competitive Equilibrium: Recursive Form

Definition. Taking as given an initial joint density $g_0(a, y)$ and an exogenous path of TFP $\{A_t\}$, a competitive equilibrium (in recursive form) comprises **functions**

$$\left\{ V_t(a, y), c_t(a, y), g_t(a, y) \right\} \quad \text{and} \quad \left\{ Y_t, \ell_t, r_t, w_t \right\}$$

such that (i) households optimize, (ii) firms optimize, (iii) markets clear, and (iv) the joint density evolves consistently with household behavior.

HJB and FOC:

$$\rho V_t(a, y_j) = u(c_t(a, y_j)) + s_t(a, y_j) \partial_a V_t(a, y_j) + \lambda_j (V_t(a, y_{-j}) - V_t(a, y_j)) + \partial_t V_t(a, y_j)$$

$$\partial_a V_t(\underline{a}, y_j) \geq u'(r_t \underline{a} + A_t y_j)$$

$$u'(c_t(a, y_j)) = \partial_a V_t(a, y_j)$$

KF:

$$\partial_t g_t(a, y_j) = -\partial_a \left[s_t(a, y_j) g_t(a, y_j) \right] - \lambda_j g_t(a, y_j) + \lambda_{-j} g_t(a, y_{-j})$$

Bond market:

$$0 = \sum_j \int a g_t(a, y_j) da$$

(We plugged in for $w_t = A_t$ and dropped goods market clearing by Walras' law)

In **continuous time**: HA models = 2 coupled PDEs! “Mean Field Games” in math.

Stationary Competitive Equilibrium

Definition. With $A_t = A$, a **stationary competitive equilibrium** comprises functions

$$\left\{ V(a, y), c(a, y), g(a, y) \right\} \quad \text{and} \quad \left\{ Y, \ell, r, w \right\}$$

such that (i) households optimize, (ii) firms optimize, (iii) markets clear, and (iv) the joint density evolves consistently with household behavior.

- Natural extension of “steady state” concept to HA economies
- Macroeconomic aggregates are constant. Distribution $g(a, y)$ is constant but households still move around as they draw idiosyncratic income shocks
- Usual notion of “steady” is: “if you start there, you stay there”

Stationary Competitive Equilibrium Conditions

$$\rho V(a, y_j) = u(c(a, y_j)) + s(a, y_j) \partial_a V(a, y_j) + \lambda_j (V(a, y_{-j}) - V(a, y_j))$$

$$\partial_a V(\underline{a}, y_j) \geq u'(r\underline{a} + Ay_j)$$

$$u'(c(a, y_j)) = \partial_a V(a, y_j)$$

$$0 = -\partial_a [s(a, y_j)g(a, y_j)] - \lambda_j g(a, y_j) + \lambda_{-j} g(a, y_{-j})$$

$$0 = \sum_j \int a g(a, y_j) da$$

Computations

Computational Advantages relative to Discrete Time

1. **Borrowing constraints** only show up in **boundary conditions**
 - FOCs always hold with “=”
2. **“Tomorrow is today”**
 - FOCs are “static”, compute by hand: $c^{-\gamma} = v_a(a, y)$
3. **Sparsity**
 - solving Bellman, distribution = inverting matrix
 - but matrices very sparse (“tridiagonal”)
 - reason: continuous time $\Rightarrow$ one step left or one step right
4. **Two birds with one stone**
 - tight link between solving (HJB) and (KF) for distribution
 - matrix in discrete (KF) is **transpose** of matrix in discrete (HJB)
 - reason: diff. operator in (KF) is **adjoint** of operator in (HJB)

Code, note, and beyond: <https://benjaminmoll.com/codes/>

Krusell and Smith (1998)

Aiyagari (1994) Model

- Introduce firms in the economy: a representative firm with production function $Y = F(K, L) = AK^\alpha L^{1-\alpha}$. Capital depreciates at rate δ_K .
- Competitive factor markets:

$$r_t = \frac{\partial F(K_t, 1)}{\partial K} - \delta_K = \alpha \frac{Y_t}{K_t} - \delta_K, w_t = \frac{\partial F(K_t, 1)}{\partial L} = (1 - \alpha) \frac{Y_t}{L_t}.$$

- Households labor productivity evolves according to an **O-U** process:

$$dz_t = \theta(\hat{z} - z_t)dt + \sigma dB_t,$$

on a bounded interval $[\underline{z}, \bar{z}]$ with $\underline{z} \geq 0$, where B_t is a Brownian motion.

- The HJB is now:

$$\rho V_t(a, z) = \frac{\partial V}{\partial t} + \max_{c \geq 0} u(c) + [w_t z + r_t a - c] \frac{\partial V}{\partial a} + \theta(\hat{z} - z) \frac{\partial V}{\partial z} + \frac{\sigma^2}{2} \frac{\partial^2 V}{\partial z^2}. \quad 14/19$$

Market clearing and the KF equation

- Aggregate productivity normalize to one: $\mathbb{E}[z_t] = 1$.
- Total assets equal **aggregate capital**:

$$\int a f_t(a, z) da dz = K_t,$$

where $f_t(a, z)$ is the **wealth-productivity density**.

- The KF equation:

$$\frac{\partial f}{\partial t} = -\frac{\partial}{\partial a}([w_t z + r_t a - c(a, z)]f_t(a, z)) - \frac{\partial}{\partial z}(\theta(\hat{z} - z)f_t(a, z)) + \frac{1}{2} \frac{\partial^2}{\partial z^2}(\sigma^2 f_t(a, z)).$$

Krusell and Smith (1998): Aiyagari model with aggregate shocks

- Consider the Aiyagari model with production function

$$Y_t = F(Z_t, K_t, L_t) = Z_t K_t^\alpha L_t^{1-\alpha} \quad (1)$$

with aggregate TFP Z following a diffusion:

$$dZ_t = \mu_z(Z_t)dt + \sigma_z(Z_t)dW_t$$

where W_t is a Brownian motion.

- Without aggregate shocks, the aggregate state of the economy $f_t(\cdot)$ is absorbed into time t : the aggregate impact on individual households is captured by $\frac{\partial v}{\partial t}$
- This is no longer the case with aggregate shocks. Aggregate state is now $(Z_t, f_t(\cdot))$.

Krusell and Smith (1998): Aiyagari model with aggregate shocks

- The HJB results in:

$$\begin{aligned}\rho V(a, z, Z, f) = & \max_{c \geq 0} u(c) + [wz + ra - c] \frac{\partial V}{\partial a} + \theta(\hat{z} - z) \frac{\partial V}{\partial z} + \frac{\sigma^2}{2} \frac{\partial^2 V}{\partial z^2} \\ & + \mu_z(Z) \frac{\partial V}{\partial Z} + \frac{\sigma_z^2(Z)}{2} \frac{\partial^2 V}{\partial Z^2} + \frac{\delta V}{\delta f} \frac{\partial f}{\partial t}.\end{aligned}$$

- The term $\frac{\delta V}{\delta Z}$ is a **functional derivative** and cannot be treated as a standard derivative.
- The KF equation:

$$\frac{\partial f}{\partial t} = -\frac{\partial}{\partial a}([w_t z + r_t a - c(a, z)]f_t(a, z)) - \frac{\partial}{\partial z}(\theta(\hat{z} - z)f_t(a, z)) + \frac{1}{2} \frac{\partial^2}{\partial z^2}(\sigma^2 f_t(a, z)).$$

- “Master equation” substitutes KFE, market clearing & belief consistency into HJB.
- We need deep learning to solve master equations globally.

Appendix

Krusell-Smith Method in Continuous Time

- **Krusell and Smith (1998)** proposed an alternative solution concept: **bounded rationality**.
- Households in the model approximate the distribution by a number of its **moments**., e.g., the mean

$$\int_0^{\infty} \int_{\underline{z}}^{\bar{z}} a f_t(a, z) da dz = K_t.$$

- The HJB becomes:

$$\begin{aligned} \rho V_t(a, z, Z, K) = & \max_{c \geq 0} u(c) + [w_t z + r_t a - c] \frac{\partial V}{\partial a} + \theta(\hat{z} - z) \frac{\partial V}{\partial z} + \frac{\sigma^2}{2} \frac{\partial^2 V}{\partial z^2} \\ & + \mu_z(Z) \frac{\partial V}{\partial Z} + \frac{\sigma_z^2(Z)}{2} \frac{\partial^2 V}{\partial Z^2} + K \mu_K(K, Z) \frac{\partial V}{\partial K} \end{aligned}$$

- Is this the same problem as in the Krusell-Smith model?

How can we compute the PLM of capital $\mu_K(K, Z)$?

Propose a parametric form of the **perceived law of motion** (PLM):

$$\mu_K(K, Z; \theta) = \theta_0 + \theta_1 K + \theta_2 KZ + \theta_3 Z.$$

Begin with an initial guess of $\theta^0 = (\theta_0^0, \theta_1^0, \theta_2^0, \theta_3^0)$. Set $n := 0$.

1. Given $\mu_K(K, Z; \theta^0)$ solve the **HJB equation** and obtain matrix **A**.
2. **Simulate** using Monte Carlo $\{Z_s\}_{s=0}^S$: $\Delta Z_s = \mu_z(Z_{s-1})\Delta t + \sigma_z(Z_{s-1})\sqrt{\Delta t}\varepsilon_s$, where $\varepsilon_s \sim \mathcal{N}(0, 1)$.
3. Compute the dynamics of the distribution using the **KF equation** and use it to obtain aggregate capital: $\int_0^\infty \int_{\underline{z}}^{\bar{z}} a f_t(a, z) da dz = K_t$.
4. Run an **OLS** $\frac{\Delta K_s}{K_s \Delta t} = \mu_K(K_s, Z_s; \theta)$ over the simulated sample $\{Z_s, K_s\}_{s=0}^S$ to update coefficients θ^{n+1} . If $\theta^{n+1} = \theta^n$ stop, otherwise go back to step 1.

Deep Learning for Search And Matching Models

(a.k.a. “DeepSAM”)

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August 25, 2026

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Introduction

- **Heterogeneity** and **aggregate shocks** are important in markets with **search frictions**. (e.g. labor and financial markets).
- Most search and matching (SAM) models with heterogeneous agents study:
 1. Deterministic steady state (e.g. Shimer-Smith '00),
 2. Aggregate fluctuations, but impose restrictions to eliminate distribution from state space (e.g. “block recursivity” in Menzio-Shi '11, Lise-Robin '17; Lagos-Rocheteau '09).
- We present SAM models as **high-dim. PDEs** with **distribution** & **agg. shocks** as states ...and develop a new deep learning method, DeepSAM, to **solve** them globally.
- We extend DeepSAM for **internal calibration with SMM** in efficient computation time.

This Paper

- Develop DeepSAM and apply to “canonical” search models with aggregate shocks:
 1. Shimer-Smith/Mortensen-Pissarides model with two-sided heterogeneity.
 2. Lise-Robin on-the-job search (OJS) model with worker bargaining power.
 3. Duffie-Garleanu-Pederson OTC model with asset and investor heterogeneity.
- High accuracy + efficient compute time for both solution and estimation.
- This talk: study unemployment and wage dynamics during business cycles and crises:
 1. Distribution feedback most important when aggregate shocks are asymmetric.
 2. Low-type worker wages are more procyclical.
 3. Low-type workers benefit more from longer expansions (“Okun’s hypothesis”).
 4. Block recursive model assumptions amplify IRFs.

Literature

- Deep learning in macro; for incomplete market heterogeneous agent models (HAM) (e.g. Maliar et al '21, Azinovic et al '22, Kahou et al '21, Han-Yang-E '21 “DeepHAM”; Fernández-Villaverde et al '19, Huang '22, Gu-Laurière-Merkel-Payne '23, among others)
 - This paper: search and matching (SAM) models.

	Distribution	Distribution impact on decisions
HAM	Asset wealth and income	Via aggregate prices
SAM	Type (productivity) of agents in two sides of matching	Via matching process with other types

- Search model with business cycle (e.g. Shimer '05, Menzio-Shi '11, Lise-Robin '17.)
 - This paper: keep distribution in the state vector.
- Integrate deep learning based solution methods with calibration and estimation (e.g. F-V et al '19, Chen et al '23, Kase et al '23, Friedl et al '23)
 - This paper: standard internal calibration practice for quantitative macro.

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Methodology

Application 1: Crisis Shock in Shimer-Smith (2000)

Application 2: Business Cycles in Lise-Robin (2017) With Worker Bargaining Power

Conclusion

Shimer-Smith Model with Aggregate Shocks

- Continuous time, infinite horizon environment.
- **Workers** $x \in [0, 1]$ with exog density $g_t^w(x)$; **Firms** $y \in [0, 1]$ with $g_t^f(y)$ by free entry:
 - Unmatched: unemployed workers get benefit b ; vacant firms pay vacancy cost c .
 - Matched: type x worker and type y firm produce output $z_t f(x, y)$.
 - z_t : follows continuous time Markov Chain (can be generalized).
 - Firms make entry decision and then draw a type y from uniform distribution $[0, 1]$. More
 - Firm exit: exogenous exit at rate $\varsigma(z)$; fraction $\sigma(z, \tilde{z})$ of firms exit when z jumps to $\tilde{z}$.
(Exiting firms/workers compensated; calibrated s.t. **no endogenous exit**.) General case
- **Meet randomly** at rate $m(\mathcal{U}_t, \mathcal{V}_t)$, $\mathcal{U}_t$ is total unemployment, $\mathcal{V}_t$ is total vacancies.
- Upon meeting, agents choose whether to accept the match:
 - Match surplus $S_t(x, y)$ divided by **Nash bargaining**: worker gets fraction β .
 - Match acceptance decision $\alpha_t(x, y) = \mathbb{1}\{S_t(x, y) \geq 0\}$. Exogenous dissolve rate $\delta(x, y, z)$.
- Equilibrium object: $g_t(x, y)$ “density” of matches $\Rightarrow$ unemployed $g_t^u(x)$, vacant $g_t^v(y)$.

Recursive Equilibrium: Agent Problems

- E.g. worker idiosyncratic state = x , Aggregate states = $(z, g(x, y))$.
Let $dg/dt = \check{\mu}^g(x, y, z, g)$ denote agents' belief about the evolution of g .
- Hamilton-Jacobi-Bellman equation (HJBE) for unemployed worker's value $V^u(x, z, g)$:

$$\begin{aligned} \rho V^u(x, z, g) = & b + \frac{m(z, g)}{\mathcal{U}(z, g)} \int \underbrace{\overbrace{\alpha(x, \tilde{y}, z, g)}^{\text{acceptance decision}} \overbrace{(V^e(x, \tilde{y}, z, g) - V^u(x, z, g))}^{\text{employed value}}}_{\text{change of value conditional on match}} \frac{g^v(\tilde{y})}{\mathcal{V}(z, g)} d\tilde{y} \\ & + \sum_{\tilde{z}} \lambda_{z\tilde{z}} (V^u(x, \tilde{z}, g^+) - V^u(x, z, g)) + \underbrace{D_g V^u(x, z, g)}_{\text{Frechet derivative: how change of } g \text{ affects } V} \cdot \underbrace{\check{\mu}^g}_{\text{Belief about } g \text{ evolution}} \end{aligned}$$

where $g^+ := (1 - \sigma(z, \tilde{z}))g$: post-jump distribution (firm exit at z jumps).

Recursive Equilibrium: Agent Problems

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where $g^+ := (1 - \sigma(z, \tilde{z}))g$: post-jump distribution (firm exit at z jumps).

+ HJBEs for employed workers V^e , vacant firms V^v , and producing firms V^p [More](#).

Recursive Equilibrium: Distribution Evolution and Definition

Given α choices, **evolution of $g_t(x, y)$** is given by Kolmogorov forward equation (KFE):

$$dg_t(x, y) = \underbrace{\mu^g(x, y, z_t, g_t) dt - \sigma(z_t, \tilde{z}) g_t(x, y) dN_t(z_t; \tilde{z})}_{\text{Jump: exogenous firm exit when } z_t \text{ jumps to } \tilde{z}}$$

$$\mu^g(x, y, z, g) = \underbrace{-(\delta(x, y, z) + \varsigma(z)) g_t(x, y)}_{\text{Outflow: breakup of matches \& firm exit}} + \underbrace{\frac{m(z, g)}{\mathcal{U}(z, g)\mathcal{V}(z, g)} \alpha(x, y, z, g) g_t^u(x) g_t^v(y)}_{\text{Inflow: creation of new matches}}$$

Recursive Equilibrium: Distribution Evolution and Definition

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A **(recursive) equilibrium** is a collection of functions $\{V^u, V^e, V^v, V^p, w, \alpha, g^f\}$ of the state variables (z, g) such that:

1. Given beliefs about the evolution of g_t , $(V^u, V^e, V^v, V^p, \alpha)$ solve the HJB equations,
2. The division of surplus satisfies Nash bargaining,
3. g^f satisfies the free entry condition [details](#), and
4. Agent beliefs about the evolution of g_t are consistent: $\check{\mu}^g = \mu^g$.

Recursive Characterization For Equilibrium Surplus

- Surplus of match $S(x, y, z, g) \equiv V^p(x, y, z, g) - V^v(y, z, g) + V^e(x, y, z, g) - V^u(x, z, g)$
- We characterize equilibrium with master equation for surplus: C.f. block recursive

$$\begin{aligned} \rho S(x, y, z, g) = & z f(x, y) - \delta(x, y, z) S(x, y, z, g) \\ & + c - (1 - \beta) \frac{m(z, g)}{\mathcal{V}(z, g; S)} \int \alpha(\tilde{x}, y, z, g) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}(z, g)} d\tilde{x} \\ & - b - \beta \frac{m(z, g)}{\mathcal{U}(z, g)} \int \alpha(x, \tilde{y}, z, g) S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\mathcal{V}(z, g; S)} d\tilde{y} \\ & + \sum_{\tilde{z}} \lambda_{z\tilde{z}} (S(x, y, \tilde{z}, (1 - \sigma(z, \tilde{z}))g) - S(x, y, z, g)) + D_g S \cdot \mu^g(z, g) \end{aligned}$$

- Kolmogorov forward equation (KFE): $dg_t(x, y) = \mu^g dt - \sigma(z_t, \tilde{z}) g_t(x, y) dN_t(z_t; \tilde{z}),$

$$\mu^g(x, y, z, g) = -(\delta(x, y, z) + \varsigma(z)) g(x, y) + \frac{m(z, g)}{\mathcal{U}(z, g) \mathcal{V}(z, g)} \alpha(x, y, z, g) g^v(y) g^u(x)$$

- High-dim PDEs with distribution in state: hard to solve with conventional methods.

Finite Type Approximation

- Approximate $g(x, y)$ on finite types: $x \in \mathcal{X} = \{x_1, \dots, x_{n_x}\}$, $y \in \mathcal{Y} = \{y_1, \dots, y_{n_y}\}$.
(Other approaches of discretization: projection and finite population [More](#))
- Finite state approximation $\Rightarrow$ analytical (approximate) KFE: $g \approx \{g_{ij}\}_{i \leq n_x, j \leq n_y}$
- Approximated master equation for surplus:

$$\begin{aligned}
 0 = \mathcal{L}^S S(x, y, z, g) = & -(\rho + \delta(x, y, z))S(x, y, z, g) + zf(x, y) + c - b \\
 & - (1 - \beta) \frac{m(z, g)}{\mathcal{V}(z, g)} \frac{1}{n_x} \sum_{i=1}^{n_x} \alpha(\tilde{x}_i, y, z, g) S(\tilde{x}_i, y, z, g) \frac{g^u(\tilde{x}_i)}{\mathcal{U}(z, g)} \\
 & - \beta \frac{m(z, g)}{\mathcal{U}(z, g)} \frac{1}{n_y} \sum_{j=1}^{n_y} \alpha(x, \tilde{y}_j, z, g) S(x, \tilde{y}_j, z, g) \frac{g^v(\tilde{y}_j)}{\mathcal{V}(z, g)} \\
 & + \sum_{\tilde{z}} \lambda_{z\tilde{z}} (S(x, y, \tilde{z}, g^+) - S(x, y, z, g)) + \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} \partial_{g_{ij}} S(x, y, z, \{g_{ij}\}_{i,j}) \mu^g(\tilde{x}_i, \tilde{y}_j, z, g)
 \end{aligned}$$

where $g_{ij}^+ = (1 - \sigma(z, \tilde{z}))g_{ij}$; μ^g from discretized KFE; smoothed acceptance

$$\alpha = (1 + e^{-\xi S(x, y, z, g)})^{-1}$$

Discrete Choice Approximation

- In the original model, acceptance is discrete: $\alpha(x, y, z, g) = \mathbb{1}\{S(x, y, z, g) \geq 0\}$
- Discrete choice $\alpha \Rightarrow S(x, y, z, g)$ non-smooth at some g (a viscosity solution)
 $\Rightarrow$ sharp decision boundaries make gradient-based NN training less stable.
- Baseline: keep master equation well defined & NN training efficient $\Rightarrow$ approximate:

$$\alpha(x, y, z, g) = \frac{1}{1 + e^{-\xi S(x, y, z, g)}}$$

Interpretation: logit choice model with utility shocks $\sim$ extreme value distribution (micro-foundation as in Caliendo et al. '19; $\xi \rightarrow \infty \Rightarrow$ discrete choice α).

- **We can also solve the original discrete choice formulation**, via “**homotopy**”: Results
 1. Train on a sequence of smooth problems with increasing curvature $\xi_1 < \dots < \xi_N$, warm-starting each stage from the previous solution;
 2. Final stage: switch to the exact discrete α , initialized from the ξ_N solution.
- Delivers accurate solutions to the discrete choice problem ($\xi = 42$ already $\approx$ discrete), ...but $\sim 10\times$ compute time: smooth α preferable if logit interpretation acceptable.

DeepSAM Algorithm for Solving the Master Equation

1. Approximate surplus by neural network $S(x, y, z, g) \approx \hat{S}(x, y, z, g; \Theta)$. Function form
2. Start with initial parameter guess Θ^0 . At iteration n with Θ^n :
 - 2.1 Generate K sample points, $Q^n = \{(x_k, y_k, z_k, \{g_{ij,k}\}_{i \leq n_x, j \leq n_y})\}_{k \leq K}$.
 - 2.2 Calculate the average mean squared error of surplus master equation on sample points:

$$L(\Theta^n, Q^n) := \frac{1}{K} \sum_{k \leq K} \left| \mathcal{L}^S \hat{S}(x_k, y_k, z_k, \{g_{ij,k}\}_{i \leq n_x, j \leq n_y}) \right|^2$$

- 2.3 Update NN parameters with stochastic gradient descent (SGD) or variants:

$$\Theta^{n+1} = \Theta^n - \zeta^n \nabla_{\Theta} L(\Theta^n, Q^n)$$

- 2.4 Repeat until $L(\Theta^n, Q^n) \leq \epsilon$ with precision threshold ϵ .

3. Once S is solved, we have α and can solve for worker and firm value functions.

Comment: Some Technical Details

- *Q. How do we choose where to sample?* Three-phase sampling & training scheme:
 - **Phase 1:** draw distributions “between” steady states (random mixtures of steady states for different fixed z); train until the loss is small.
 - **Phase 2:** move to simulation based (“ergodic”) sampling under the current $\hat{S}$, with a training/validation split, and periodic resimulation as $\hat{S}$ improves.
 - **Phase 3:** report out-of-sample loss on a freshly simulated evaluation set. [More](#)
- *Q. Why global solution?* Non-linear approximation across broad state space.
- *Q. Which equilibrium concept?*
 - We solve for a rational expectations equilibrium
 $\Rightarrow$ requires solving the master equation where $\mu^g = \check{\mu}^g$ on and off equilibrium path.
 - Different to “reinforcement learning” where agents optimize in response to simulations
 $\Rightarrow \mu^g \neq \check{\mu}^g$ does not necessarily hold, especially off equilibrium path. [More](#)

Neural networks do not train in seconds.

So how can we choose structural parameters to match data?

DeepSAM for Internal Calibration with Simulated Method of Moments

- So far, we have solved:

$$0 = \mathcal{L}^S S(x, y, z, \underline{g})$$

- For internal calibration, we include economic parameters Ψ as pseudo states and solve extended master equation:

$$0 = \mathcal{L}^S S(x, y, z, \underline{g}, \Psi)$$

- Simulate the model under different structural parameter vectors $\{\Psi_l\}_l$ and ... fit a surrogate function mapping structural parameters to simulated moments.
- Use surrogate function to find the parameters that match data moments.

More

Extensions

- On-the-job search with a parameterized bargaining protocol nesting: [More](#)
 - Bertrand competition between firms (Postel-Vinay and Robin, 2002),
 - sequential auctions (Cahuc, Postel-Vinay and Robin, 2006), and
 - Nash bargaining over incremental surplus from moving (Pissarides, 1994).
- Match specific idiosyncratic productivity shocks. [More](#)
- Endogenous separation. [More](#)
- Variations on free entry and exit. [More](#)
- Agent type switching. [More](#)
- Asset trade rather than worker-firm production (e.g., OTC search markets). [More](#)
- Non-transferable utility. [More](#)

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Methodology

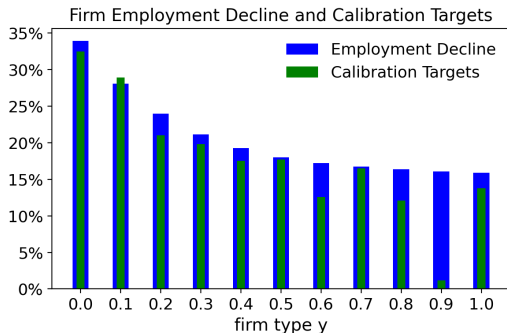
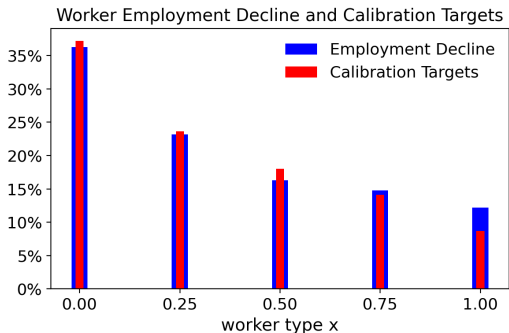
Application 1: Crisis Shock in Shimer-Smith (2000)

Application 2: Business Cycles in Lise-Robin (2017) With Worker Bargaining Power

Conclusion

Crisis Shock in Shimer-Smith (2000)

- We test the solution method for the baseline model
... with a recession where the job separation rate significantly increases.
- We calibrate the crisis separation rate $\delta_D(x, y)$ to match the heterogeneous employment declines during COVID-19 for different workers/firms (Cajner et al., 2020). [More](#)



- Worker and firm types: $(n_x, n_y) = (5, 11) \Rightarrow 58$ dimensional PDE for $S(x, y, z, g)$

Numerical Accuracy of Solution Technique

Technically, our method achieves high numerical accuracy using a number of measures:

- **Small numerical error.** Use DeepSAM to solve the problem (58 dimensional PDE), and compute loss on simulated paths not used in training. Evaluation RMSE: 1.7×10^{-3} (training RMSE: 9.2×10^{-4}). [More](#)
- **Verification on models with known solution.** Use DeepSAM to solve model without aggregate shocks (57 dimensional PDE) and obtain steady state solution. Compare to steady state solution from conventional methods. RMSE Difference: 2.4×10^{-3} . [More](#)

Q1. Does Distribution Feedback Matter? Evidence From COVID-19

- Aggregate dynamics **with** and **without** distribution feedback to agent decisions:

Full dynamics: $\mu_t^g(x, y) = -(\delta(x, y, z_t) + \varsigma(z_t))g_t(x, y) + \frac{m_t(z, g_t)}{\mathcal{U}_t \mathcal{V}_t} \alpha(x, y, z_t, \mathbf{g}_t) g_t^u(x) g_t^v(y)$

No distribution feedback: $\mu_t^g(x, y) = -(\delta(x, y, z_t) + \varsigma(z_t))g_t(x, y) + \frac{m_t(z, g_t)}{\mathcal{U}_t \mathcal{V}_t} \alpha(x, y, z_t, \mathbf{g}^{\text{ergodic}}) g_t^u(x) g_t^v(y)$

(Both share the same exogenous exit jumps at z_t changes; $\mathbf{g}^{\text{ergodic}}$: ergodic mean of $\mathbf{g}_t$.)

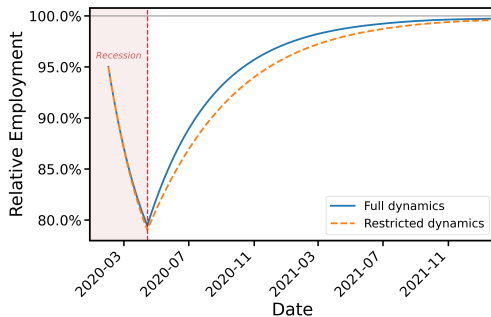
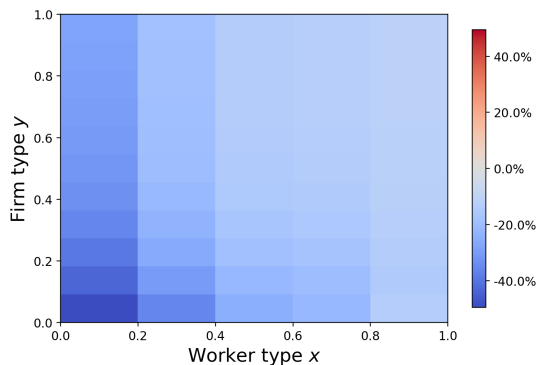


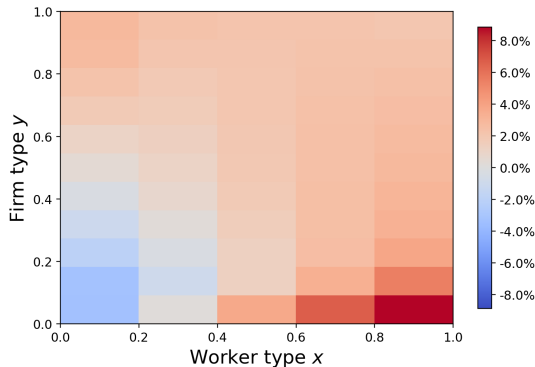
Figure: Employment change after the COVID-19 shock

A1. Mechanism: Asymmetric Shock Makes Agents “Less Picky”

Intuition: COVID-19 leads to relatively more low-type unemployed workers and firms
 $\Rightarrow$ workers and firms are less willing to wait for a good match.



(a) Distribution difference: after COVID-19 shock relative to the ergodic mean.



(b) Acceptance difference: after COVID-19 shock relative to the ergodic mean.

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Application 1: Crisis Shock in Shimer-Smith (2000)

Application 2: Business Cycles in Lise-Robin (2017) With Worker Bargaining Power

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Solving and Estimating OJS Model

- Lise-Robin (2017) with worker bargaining power $\beta > 0$ [Model details](#)
- Discretization: $(n_x, n_y) = (7, 8) \Rightarrow S(x, y, z, g)$ is 59-dimensional.

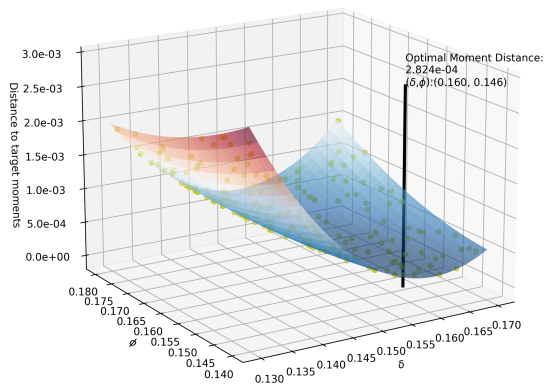
Solving and Estimating OJS Model

- Lise-Robin (2017) with worker bargaining power $\beta > 0$ Model details
- Discretization: $(n_x, n_y) = (7, 8) \Rightarrow S(x, y, z, g)$ is 59-dimensional.
- Internal calibration for $\{\beta, \kappa, c, \delta, \phi, \eta, b\}$: solve the model over economic parameter space, and simulate across 10,000 parameter combinations for simulated method of moments.

	Solving Model For a Parameter Set	Solving Model with Structural Parameters as Pseudo-states	Training Surrogate Model	Simulated Method of Moments	Entire Estimation
RMSE Loss	9.97×10^{-4}	2.9×10^{-3}	7.82×10^{-4}	1.24×10^{-4}	-
Time	40min	2h 47min	23min	0.4min	3h 10min

Moments	$\mathbb{E}[U]$	$\mathbb{E}[V]$	$\mathbb{E}[EE]$	$\mathbb{E}[UE]$	$\mathbb{E}[EU]$
Data	0.058	0.037	0.025	0.421	0.025
Model	0.057	0.036	0.026	0.448	0.027

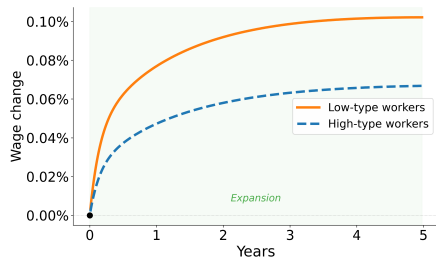
Estimation of OJS Model: Visualization in 2D



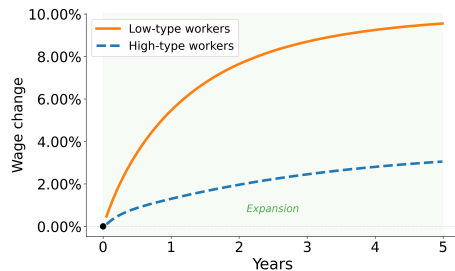
Distance to target moments over separation rate δ and OJS intensity ϕ .

Q2. Are wage dynamics heterogeneous across distribution? A2. Yes

- In Lise-Robin: “wages cannot be solved for exactly... need to solve worker values where the distribution of workers across jobs is a state variable.”
- DeepSAM can solve wage dynamics with rich heterogeneity.
- Low-type worker wages more procyclical — under both protocols $\chi = 0$ and $\chi = 1$.



(a) IRF to positive shocks ($\chi = 0$)



(b) IRF to positive shocks ($\chi = 1$)

Figure: Wage change after z shocks for $\chi = 0$, $\chi = 1$. Low/middle/high: lowest, middle, highest of $n_x = 7$ worker types.

Q3. Who benefits more over a longer expansion? A3. Low Types

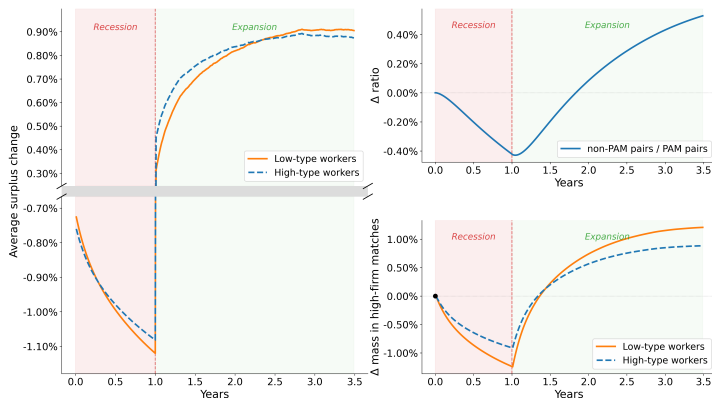


Figure: Left: Δ avg. surplus by worker type. Top right: non-PAM/PAM mass ratio. Bottom right: Δ employment at firms with $y > 1/2$.

- Low-type workers overtake high types in surplus growth ~ 2 years into the expansion.
- Mechanism: sorting weakens over time in expansions — high-type workers become scarce, so high-type firms become less picky and hire more low-type workers.

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Conclusion

- We develop an integrated global solution and estimation method, DeepSAM, to search and matching models with heterogeneity and aggregate shocks.
- We apply DeepSAM to three general setups in labor and financial search models (without simplification assumptions). [OTC model details](#)
- The method is accurate, solves new variables (e.g. wage), and generates novel economic insights.
- A foundational tool for a large literature with more applications:
 - Richer models in labor, financial, and money search, combined with rich micro data.
 - Spatial and network models with aggregate uncertainty (similar math structure).

Thank You!

Approximate S by Neural Network (Feed Forward, Fully Connected)

- Let $\omega = (x, y, z, g)$. We approximate surplus $S(\omega)$ by neural network with form:

$$\mathbf{h}^{(1)} = \phi^{(1)}(W^{(1)}\omega + \mathbf{b}^{(1)}) \quad \dots \text{Hidden layer 1}$$

$$\mathbf{h}^{(2)} = \phi^{(2)}(W^{(2)}\mathbf{h}^{(1)} + \mathbf{b}^{(2)}) \quad \dots \text{Hidden layer 2}$$

$$\vdots$$

$$\mathbf{h}^{(H)} = \phi^{(H)}(W^{(H)}\mathbf{h}^{(H-1)} + \mathbf{b}^{(H)}) \quad \dots \text{Hidden layer H}$$

$$S = \sigma(\mathbf{h}^{(H)}) \quad \dots \text{Surplus}$$

- Terminology (our parameter choices are in blue):
 - H : is the number of *hidden layers*, ($H = 4; 5$ for estimation with pseudo-states)
 - Length of vector $\mathbf{h}^{(i)}$: number of *neurons* in hidden layer i , ($Length = 50; 100$ for estimation)
 - $\phi^{(i)}$: is the *activation function* for hidden layer i , ($\phi^i = \tanh$)
 - σ : is the *activation function* for the final layer, ($\sigma = \tanh$)
 - $\Theta = (W^1, \dots, W^{(H)}, b^{(1)}, \dots, b^{(H)})$ are the *parameters*,

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Methodology

Labor Search Model

Entry and Exit

On-The-Job Search Model

On-The-Job Search Model with Idiosyncratic Shocks

OTC Market

Non-transferable Utility (NTU)

Free Entry Condition

- Firms make entry decision and then draw type y from uniform distribution $[0, 1]$:

$$0 = \mathbb{E}[V_t^v] = \int V^v(\tilde{y}, z, g) d\tilde{y}. \quad (1)$$

- As the matching function is homothetic $\frac{m(z_t, g_t)}{\mathcal{V}_t} = \hat{m}\left(\frac{\mathcal{V}_t}{\mathcal{U}_t}\right)$, combining free entry condition with HJB equation for V^v gives:

$$\hat{m}\left(\frac{\mathcal{V}_t}{\mathcal{U}_t}\right) = \frac{c}{\int \int \alpha(\tilde{x}, \tilde{y}) \frac{g_t^u(\tilde{x})}{\mathcal{U}_t} (1 - \beta) S_t(\tilde{x}, \tilde{y}) d\tilde{x} d\tilde{y}} \Rightarrow \mathcal{V}_t = \mathcal{U}_t \hat{m}^{-1}(\dots) \quad (2)$$

where $g_t^u = g_t^w - \int g_t^m(x, y) dy$ and so the RHS can be computed from g_t^m and S_t .

- $g_t^f(y) = \mathcal{V}_t + \mathcal{P}_t$ for all y (uniform), where $\mathcal{V}_t, \mathcal{P}_t$ can be expressed in terms of g and S .
- Timing at a jump in z : z changes $\rightarrow$ exogenous firm exit (σ) $\rightarrow g_t^f$ adjusts to restore free entry.
- With free entry condition, the master equation expression for surplus takes the same form as without free entry, but with different expressions of $g^f(y)$.

Recursive Equilibrium Part II: Other Equations

- Hamilton-Jacobi-Bellman equation (HJBE) for employed worker's value $V^e(x, y, z, g)$:

$$\begin{aligned}\rho V^e(x, y, z, g) = & w(x, y, z, g) + \delta(x, y, z) (V^u(x, z, g) - V^e(x, y, z, g)) \\ & + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (V^e(x, y, \tilde{z}, g^+) - V^e(x, y, z, g)) + D_g V^e(x, y, z, g) \cdot \check{\mu}^g\end{aligned}$$

- HJBE for a vacant firm's value $V^v(y, z, g)$:

$$\begin{aligned}\rho V^v(y, z, g) = & -c + \frac{m(z, g)}{\mathcal{V}(z, g)} \int \alpha(\tilde{x}, y, z, g) (V^p(\tilde{x}, y, z, g) - V^v(y, z, g)) \frac{g^u(\tilde{x})}{\mathcal{U}(z, g)} d\tilde{x} \\ & + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (V^v(y, \tilde{z}, g^+) - V^v(y, z, g)) + D_g V^v(y, z, g) \cdot \check{\mu}^g\end{aligned}$$

- HJBE for a producing firm's value $V^p(x, y, z, g)$:

$$\begin{aligned}\rho V^p(x, y, z, g) = & z f(x, y) - w(x, y, z, g) + \delta(x, y, z) (V^v(y, z, g) - V^p(x, y, z, g)) \\ & + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (V^p(x, y, \tilde{z}, g^+) - V^p(x, y, z, g)) + D_g V^p(x, y, z, g) \cdot \check{\mu}^g\end{aligned}$$

- $g^+ := (1 - \sigma(z, \tilde{z}))g$: post-jump distribution. No direct ς/σ terms: full compensation for exit.

Comparison to Other Heterogeneous Agent Search Models

- **Lise-Robin '17:** sets $\beta = 0$ (and other conditions, including Postal-Vinay Robin style Bertrand competition for workers searching on-the-job)

$$S(x, y, z, \textcolor{red}{g}) = S(x, y, z), \quad \alpha(x, y, z, \textcolor{red}{g}) = \alpha(x, y, z)$$

- **Menzio-Shi '11:** competitive search (directed across a collection of sub-markets):

$$S(x, y, z, \textcolor{red}{g}) = S(x, y, z)$$

- We look for a solution for S and α in terms of the distribution g .

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Modification 1: Finite Type Approximation

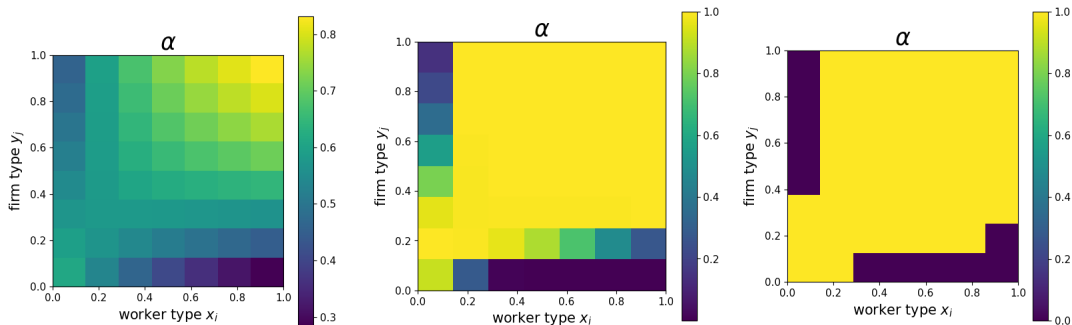
- Approximate $g(x, y)$ on finite types: $x \in \mathcal{X} = \{x_1, \dots, x_{n_x}\}$, $y \in \mathcal{Y} = \{y_1, \dots, y_{n_y}\}$.
- Finite state approximation $\Rightarrow$ analytical (approximate) KFE: $g \approx \{g_{ij}\}_{i \leq n_x, j \leq n_y}$
- Approximated master equation for surplus:

$$\begin{aligned} 0 = \mathcal{L}^S S(x, y, z, g) = & -(\rho + \delta(x, y, z))S(x, y, z, g) + zf(x, y) + c - b \\ & - (1 - \beta) \frac{m(z, g)}{\mathcal{V}(z, g)} \frac{1}{n_x} \sum_{i=1}^{n_x} \alpha(\tilde{x}_i, y, z, g) S(\tilde{x}_i, y, z, g) \frac{g^u(\tilde{x}_i)}{\mathcal{U}(z, g)} \\ & - \beta \frac{m(z, g)}{\mathcal{U}(z, g)} \frac{1}{n_y} \sum_{j=1}^{n_y} \alpha(x, \tilde{y}_j, z, g) S(x, \tilde{y}_j, z, g) \frac{g^v(\tilde{y}_j)}{\mathcal{V}(z, g)} \\ & + \sum_{\tilde{z}} \lambda_{z\tilde{z}} (S(x, y, \tilde{z}, g^+) - S(x, y, z, g)) + \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} \partial_{g_{ij}} S(x, y, z, \{g_{ij}\}_{i,j}) \mu^g(\tilde{x}_i, \tilde{y}_j, z, g) \end{aligned}$$

where $g_{ij}^+ = (1 - \sigma(z, \tilde{z}))g_{ij}$ is the post-jump match distribution.

Modification 2: Solving the Exact Discrete Choice by Homotopy

- Acceptance α at steady state g : smooth ($\xi = 2$, $\xi = 42$) vs. exact discrete rule:



- High curvature ($\xi = 42$) already closely approximates the exact discrete solution.
- Homotopy training is stable: loss declines within each ξ stage, jumps at stage switches, and the final discrete stage converges from the high- ξ warm start.
- C.f. perturbation approaches (Reiter '09, Bilal '23), which rely on differentiability.

Finite Dimensional “Distribution” Approximation ($a = \text{idio. state}$)

	Finite Population	Discrete State	Projection
Params $\hat{\varphi}$	Agent states $\hat{\varphi}_t = \{(a_t^i)\}_{i \leq N}$	Masses on grid $\hat{\varphi}_{i,t}, \forall (a^i)_{i \leq N}$	Basis coefficients $\hat{\varphi}_{i,t}, \forall b_i(a) _{i \leq N}$
Dist. approx.	$\frac{1}{N} \sum_{i=1}^N \delta_{(a_t^i)}$	$\sum_{i=1}^N \hat{\varphi}_{i,t} \delta_{(a^i)}$	$\sum_{i=0}^N \hat{\varphi}_{i,t} b_i(a) \approx \mu_g \left(a, z, \sum_{i=1}^N \hat{\varphi}_{i,t} b_i \right)$
KFE approx. ($\mu^{\hat{\varphi}}$)	Evolution of other agents' states	Evolution of mass between grid points (e.g. finite diff.)	Evolution of projection coefficients (least squares)

Finite population works well for Walrasian markets; less well for search

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Sampling and Training Scheme for the Surplus Network

- Sampling (x, y, z) is easy (low-dim): multinomial sampler on the discretized support. Sampling the distribution (g_{ij}) is high-dim $\Rightarrow$ three-phase scheme:
- *Phase 1 — mixed steady-state sampler:*
 1. Solve for the steady state g_m^s for each fixed aggregate state $s \in \{L, H, D\}$.
 2. Draw random convex combinations $g = \sum_s w_s g_m^s$: draw a dominant state s^* with stationary prob. π_s , set $w_{s^*} \sim \text{Beta}(\alpha, 1)$ with $\alpha \gg 1$, split the rest $\propto \pi$.
 3. Train on these samples until the master equation loss falls below a threshold.
- *Phase 2 — ergodic simulation sampler, inner-outer training:*
 1. Simulate (z_t, g_t) forward with the KFE under the current $\hat{S}$; discard burn-in $\Rightarrow$ ergodic pool of (g, z) draws, split into training/validation sets $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}$.
 2. *Inner loop:* SGD on minibatches from $\mathcal{D}_{\text{train}}$; track validation loss on $\mathcal{D}_{\text{val}}$ and keep the best weights Θ^* (validation-based model selection).
 3. *Outer loop:* resimulate under $\Theta^* \Rightarrow$ fresh $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}$; repeat until validation loss stabilizes.
- *Phase 3 — out-of-sample evaluation:* simulate a new evaluation set $\mathcal{D}_{\text{eval}}$ after training; report the evaluation loss (the RMSE in our accuracy results).

Comment 2: Different NN Techniques $\Leftrightarrow$ Different Eqm Concepts

Borrowing from Sargent (2008): [Back](#)

- **Rational expectations eqm** ([this paper](#); $\mu^g = \check{\mu}^g$):
 - Agent “perceived law of motion (PLM)” is the *true law of motion* if agents follow PLM
 - True *on and off the equilibrium path* $\Rightarrow$ requires solving the master equation.
- **Self-confirming eqm** ([some model-based reinforcement learning](#), e.g. DeepHAM)
 - PLM is *best statistical fit to data generated by economy* where agents follow that PLM.
 - So, agents would never reject a misspecification test
 - but may have off-equilibrium beliefs that differ from rational expectations ($\mu^g \neq \check{\mu}^g$)
 - $\Rightarrow$ can be solved by updating reaction functions.
- **Misspecified belief eqm**
 - Agents have a “perceived law of motion” for the equilibrium variables in the economy
 - ...that *satisfies a fixed parametric form* but doesn't satisfy a misspecification test.

Calibration and Estimation Challenges

- Let $\Psi \in \Omega^\Psi$ be the structural parameters to be calibrated internally.
- Let $\check{\varphi} = (\check{\varphi}_1, \dots, \check{\varphi}_N)$ be the $N \times 1$ data moments that we want to match.
- Let $\phi(\Psi) = (\phi_1(\Psi), \dots, \phi_N(\Psi))$ be the corresponding simulated model moments.
- **Computation challenge:** need to solve $\phi(\Psi)$ for many structural parameters Ψ .
- **Solution:** include Ψ as a pseudo state vector and solve extended master equation:

$$0 = \mathcal{L}^S S(x, y, z, \underline{g}, \Psi)$$

- Approximate extended surplus function by NN: $S(x, y, z, g, \Psi) \approx \hat{S}(x, y, z, g, \Psi; \Theta)$.
- Fit $\hat{S}$ using same approach as before but sampling over states & structural parameters.

Simulated Method of Moments

- Fit surrogate neural network $\hat{\Phi}$ mapping structural parameters to simulated moments:

$$\Psi \mapsto \hat{\Phi}(\Psi; \Theta^\Phi) = \left(\hat{\phi}_1(\Psi; \Theta^\Phi), \dots, \hat{\phi}_N(\Psi; \Theta^\Phi) \right), \quad \Theta^\Phi \text{ are NN parameters}$$

- Simulate the model under many different structural parameter vectors $\{\Psi_l\}_l$
 - Compute the resulting model moments $\{(\phi_1(\Psi_l), \dots, \phi_N(\Psi_l))\}_l$
 - Fit neural network parameters Θ^Φ to approx relationship $\hat{\Phi}(\Psi; \Theta^\Phi)$ using simulations.
- **Think:** surrogate NN has incorporated the economic structure implicitly into Θ^Φ
 - With $\hat{\Phi}(\Psi; \Theta^\Phi)$, we can find the parameters to match data moments:
(i.e. perform simulated method of moments)

$$\Psi^* = \arg \min_{\Psi} \sum_{i=1}^N \omega_i \left(\frac{\check{\varphi}_i - \hat{\phi}_i(\Psi; \Theta^\Phi)}{\check{\varphi}_i} \right)^2.$$

Calibration and Estimation

1. Fit extended surplus function by NN: $S(x, y, z, g, \Psi) \approx \hat{S}(x, y, z, g, \Psi; \Theta)$.
Start with initial parameter guess Θ^0 . At iteration n with Θ^n :

1.1 Generate K sample points, $Q^n = \{(x_k, y_k, z_k, \{g_{ij,k}\}_{i \leq n_x, j \leq n_y}, \Psi_k)\}_{k \leq K}$.

1.2 Calculate the average mean squared error of surplus master equation on sample points:

$$L(\Theta^n, Q^n) := \frac{1}{K} \sum_{k \leq K} \left| \mathcal{L}^S \hat{S}(x_k, y_k, z_k, \{g_{ij,k}\}_{i \leq n_x, j \leq n_y}, \Psi_k) \right|^2$$

1.3 Update NN parameters with stochastic gradient descent (SGD) method.

2. Fit surrogate NN mapping structural parameters to moments: $\hat{\Phi}(\Psi; \Theta^\Phi)$.
3. Use surrogate NN to find parameters that match data moments.

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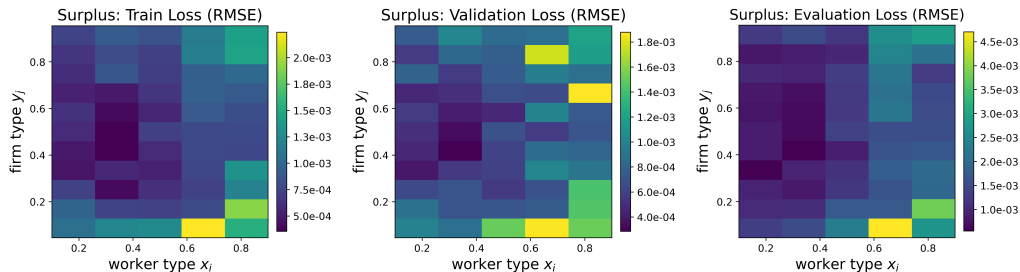
On-The-Job Search Model with Idiosyncratic Shocks

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Numerical Performance: Accuracy I [back](#)

- RMSE of the surplus master equation over the ergodic mean, computed on: (a) training samples (last inner loop), (b) validation samples (last outer loop), (c) a fresh simulated *evaluation* set.



- Average RMSEs: 9.2×10^{-4} (training), 9.1×10^{-4} (validation), 1.7×10^{-3} (evaluation)
 $\Rightarrow$ accurate approximation of S in the neighborhood of the ergodic mean.

Numerical Performance: Accuracy II

- Compare **steady state solution without aggregate shocks** to solution using conventional methods.

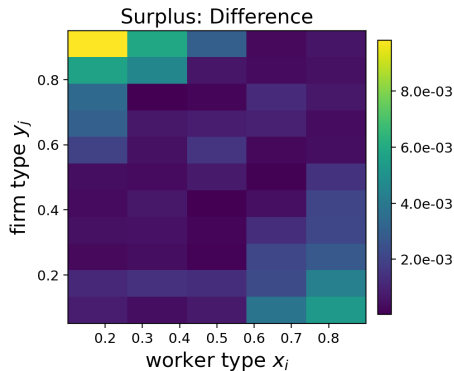
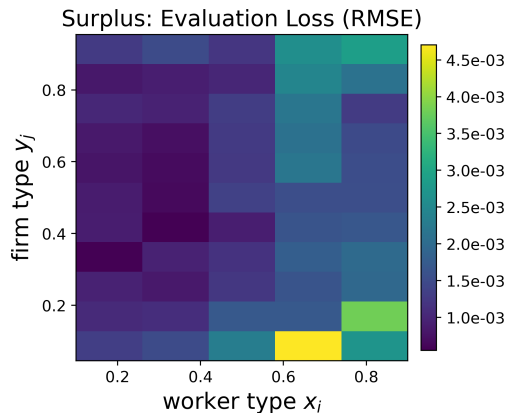


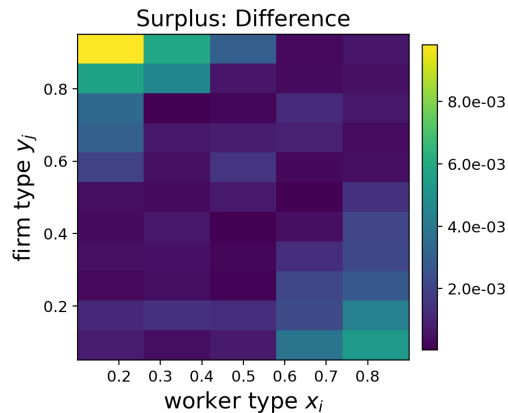
Figure: Absolute difference of S from conventional solution at DSS (order 10^{-3} ; max $S \approx 0.65$).

Calibration of Model with Aggregate Shocks [Back](#)

Parameter	Interpretation	Value	Target/Source
ρ	Discount rate	0.05	Interest rate
ξ	Extreme value for α choice	2.0	
$f(x, y)$	Production for match (x, y)	$0.6 + 0.4 (\sqrt{x} + \sqrt{y})^2$	Hagedorn et al. (2017)
β	Surplus division factor	0.72	Shimer (2005)
$m(\mathcal{U}, \mathcal{V})$	Matching function	$\kappa \mathcal{U}^\nu \mathcal{V}^{1-\nu}$	Hagedorn et al. (2017)
ν	Elasticity in meeting function	0.5	Hagedorn et al. (2017)
κ	Scale for meeting function	6.4	Unemployment rate
b	Worker unemployment benefit	1.0	Unemployment rate
c	Entry cost	2.40	Steady state $\mathcal{V}/\mathcal{U} = 1$
Steady State:			
$\bar{z}$	Steady state TFP	1	Shimer (2005)
$\bar{\delta}$	Steady state separation rate	0.2	BLS job tenure 5 years
$\bar{\varsigma}$	Steady state exogenous exit rate	0.01	No endogenous exit
Exogenous Aggregate Shock Process:			
A	TFP level	1.0	
$\varsigma_D, \varsigma_L, \varsigma_H$	Exogenous exit rates	0.63, 0.01, 0.01	No endogenous exit
δ_L, δ_H	Separation rates	0.22, 0.18	Shimer (2005)
$\delta_D(x, y)$	Separation at crisis state	0.23 to 3.9	Cajner et al. (2020)
λ_z	Poisson transition probabilities	0.4, 0.001, 0.0995	Shimer (2005)
$\sigma(z, \check{z})$	Exogenous exit at Poisson jumps	0 to 0.05	No endogenous exit
n_x	Discretization of worker types	5	Cajner et al. (2020)
n_y	Discretization of firm types	11	Cajner et al. (2020)



(a) Model with aggregate shock:
evaluation RMSE across the (x, y) grid



(b) Model without aggregate shock:
difference from conventional solution

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Different Versions of Entry and Exit

1. **Baseline (main text):** free entry + exogenous exit at rate $\varsigma(z)$ and fraction $\sigma(z, \tilde{z})$ at z jumps, with *full compensation* for exit (g^f is a jump variable).
Calibrated s.t. ς, σ large enough $\Rightarrow$ **no endogenous exit on equilibrium path.**
2. **General case:** endogenous firm exit + partial compensation (fraction $\tau \in [0, 1]$ of lost surplus) (g^f jump variable). $\tau = 1$ & no endogenous exit $\Rightarrow$ baseline.
3. **Short-lived vacancies** (as in Lise-Robin '17; used in our OJS model): unfilled vacancies destroyed each instant, $V^v = 0$; no compensation; $\varsigma = \sigma = 0$.
4. **Slow entry and exit** (*mass of firms $\mathcal{F}$ is a state variable*).

1. Baseline: Free Entry, Exogenous Compensated Exit Back

- KFE: drift includes firm exit at rate $\varsigma(z)$; exogenous jump exit when z changes:

$$dg_t(x, y) = \mu^g(x, y, z_t, g_t) dt - \sigma(z_t, \tilde{z}) g_t(x, y) dN_t(z_t; \tilde{z})$$

$$\mu^g(x, y, z, g) = -(\delta(x, y, z) + \varsigma(z))g(x, y) + \frac{m(z, g)}{\mathcal{U}(z, g)\mathcal{V}(z, g)}\alpha(x, y, z, g)g^v(y)g^u(x)$$

- Master equation: no direct ς/σ loss terms (exit compensated); jump at post-jump dist.:

$$\begin{aligned} \rho S(x, y, z, g) = & z f(x, y) - \delta(x, y, z) S(x, y, z, g) + c - b \\ & - (1 - \beta) \frac{m(z, g)}{\mathcal{V}(z, g; S)} \int \alpha(\tilde{x}, y, z, g) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}(z, g)} d\tilde{x} \\ & - \beta \frac{m(z, g)}{\mathcal{U}(z, g)} \int \alpha(x, \tilde{y}, z, g) S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\mathcal{V}(z, g; S)} d\tilde{y} \\ & + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (S(x, y, \tilde{z}, (1 - \sigma(z, \tilde{z}))g) - S(x, y, z, g)) + D_g S \cdot \mu^g \end{aligned}$$

- Firm distribution g^f jumps to ensure free entry condition is satisfied:

$$0 = \mathbb{E}[V_t^v] = \int V^v(\tilde{y}, z, g) d\tilde{y}.$$

2. General Case: Endogenous Exit and Partial Compensation Back

- If free entry would require *negative* entry, firms are randomly forced to exit: at rate $\alpha_\mu^g(z, g)$ in the drift (indicator $\mathbf{1}_{\mu^F}$) or with prob. $\alpha_\sigma^g(z, \check{z}, g)$ at z jumps (indicator $\mathbf{1}_{\sigma^F}$), chosen so the free entry condition keeps holding.
- Exiting firms (workers) receive a transfer for fraction τ of their lost surplus.
- Master equation with endogenous exit and partial compensation:

$$\begin{aligned}
 \rho S(x, y, z, g) = & z f(x, y) - \delta(x, y, z) S(x, y, z, g) - (\varsigma(z) + \mathbf{1}_{\mu^F} \alpha_\mu^g) (1 - \tau) S(x, y, z, g) \\
 & + c - (1 - \beta) \frac{m(z, g)}{\mathcal{V}(z, g; S)} \int \alpha(\tilde{x}, y, z, g) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}(z, g)} d\tilde{x} \\
 & - b - \beta \frac{m(z, g)}{\mathcal{U}(z, g)} \int \alpha(x, \tilde{y}, z, g) S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\mathcal{V}(z, g; S)} d\tilde{y} \\
 & + \sum_{\check{z} \neq z} \lambda(z, \check{z}) \left\{ S(x, y, \check{z}, g + \gamma^g) - S(x, y, z, g) \right. \\
 & \left. - ((1 - \sigma) \mathbf{1}_{\sigma^F} \alpha_\sigma^g + \sigma) (1 - \tau) S(x, y, \check{z}, g + \gamma^g) \right\} + \langle D_g S, \mu^g \rangle
 \end{aligned}$$

where $\gamma^g = -(\sigma + \mathbf{1}_{\sigma^F} \alpha_\sigma^g)g$ is the jump in g ; μ^g includes exit rate $\varsigma + \mathbf{1}_{\mu^F} \alpha_\mu^g$.

- $\tau = 1$ and $\mathbf{1}_{\mu^F} = \mathbf{1}_{\sigma^F} = 0$ recovers the baseline master equation.

3. Slow Entry and Exit Back

- Vacancies long-lived; firms exit at exogenous rate ς (no jump exit, $\sigma = 0$); entrants draw type y from density $\bar{h}^f(y) = 1$; entry at rate $\psi \alpha^f$:

$$dg_t^f(y) = \underbrace{\psi \alpha^f(z, g, \mathcal{F}) \bar{h}^f(y) dt}_{\text{entry inflow}} - \underbrace{\varsigma g_t^f(y) dt}_{\text{exit outflow}}$$

$$\alpha^f(z, g, \mathcal{F}) = \mathbb{1} \left\{ \int V^v(\tilde{y}, z, g, \mathcal{F}) \bar{h}^f(\tilde{y}) d\tilde{y} > 0 \right\}$$

- Firm density stays uniform $\Rightarrow$ only the *mass* of firms $\mathcal{F}_t$ is an extra state:

$$d\mathcal{F}_t = \psi \alpha^f(z, g, \mathcal{F}) dt - \varsigma \mathcal{F}_t dt, \quad \mathcal{V}_t = \mathcal{F}_t - \mathcal{P}_t$$

- Surplus function has an additional state (and we need to solve for V^v): $S(x, y, z, g, \mathcal{F})$, with the master equation gaining the term $\partial_{\mathcal{F}} S(x, y, z, g, \mathcal{F}) \mu^{\mathcal{F}}(z, g, \mathcal{F})$.
- Limits: $\psi \rightarrow \infty$ recovers free entry (g^f jump variable); $\psi \rightarrow 0$ shuts down entry.

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On-The-Job Search (OJS): Additional Environment Features

- Same worker types, firm types, and production function.
- Now all workers search; meetings $m(\mathcal{W}_t, \mathcal{V}_t)$; total search effort $\mathcal{W}_t := \mathcal{U}_t + \phi \mathcal{E}_t$.
Meeting rates: $\mathcal{M}_t^u = \frac{m_t}{\mathcal{W}_t}$ (unemployed), $\mathcal{M}_t^e = \phi \frac{m_t}{\mathcal{W}_t}$ (employed), $\mathcal{M}_t^v = \frac{m_t}{\mathcal{V}_t}$ (vacancy).
- Vacant $\tilde{y}$ -firm meets unemployed x -worker: Nash bargaining with worker weight β .
- Poaching firm $\tilde{y}$ meets worker in (x, y) match: **parameterized bargaining protocol**:
 - Worker moves iff $S_t(x, \tilde{y}) \geq S_t(x, y) \geq 0$; gets share β of incremental surplus $S_t(x, \tilde{y}) - S_t(x, y)$
... and share $\chi \in [0, 1]$ of the incumbent firm's surplus.
 - Nests: Bertrand competition ($\chi = 1, \beta = 0$; PVR '02), sequential auctions ($\chi = 1, \beta > 0$; CPVR '06), incremental Nash bargaining ($\chi = 0$; Pissarides '94).
 - χ does *not* enter the surplus master equation (only wages) $\Rightarrow$ same S for all χ .
- Endogenous separation at rate η when $S_t(x, y) < 0$: $\alpha_t^b(x, y) = 1 - \alpha_t(x, y)$.
- Short-lived vacancies (Lise-Robin '17): unfilled vacancies destroyed each instant $\Rightarrow V^v = 0$; free entry on the *flow* value of a vacancy; $\varsigma = \sigma = 0$.

Recursive Characterization For Equilibrium Surplus (OJS)

- Can characterize equilibrium with the master equation for the surplus:

$$\begin{aligned}\rho S(x, y, z, g) &= z f(x, y) - b - (\delta(x, y, z) + \eta \alpha^b(x, y, z, g)) S(x, y, z, g) \\ &+ \mathcal{M}^e \int \alpha^e(x, y, \tilde{y}, z, g) \beta(S(x, \tilde{y}, z, g) - S(x, y, z, g)) \frac{g^v(\tilde{y})}{\nu} d\tilde{y} \\ &- \mathcal{M}^u \int \alpha(x, \tilde{y}, z, g) \beta S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\nu} d\tilde{y} \\ &+ \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (S(x, y, \tilde{z}, g) - S(x, y, z, g)) + D_g S(x, y, z, g) \cdot \mu^g(z, g)\end{aligned}$$

where the on-the-job acceptance and endogenous breakup indicators are:

$$\alpha^e(x, y, \tilde{y}, z, g) := \mathbb{1}\{S(x, \tilde{y}, z, g) \geq S(x, y, z, g) \text{ and } S(x, \tilde{y}, z, g) \geq 0\}, \quad \alpha^b = 1 - \alpha$$

- Simpler than baseline: $V^v = 0$ (short-lived vacancies) $\Rightarrow$ no firm opportunity cost term.

$\beta = 0$ kills both g -dependent terms $\Rightarrow$ block recursivity as in Lise-Robin '17.

OJS: Free Entry Condition

- The *flow* value of creating a vacancy y : rents from hiring unemployed workers and poaching employed workers ($\mathcal{C}^u = \frac{\mathcal{U}}{\mathcal{W}}$, $\mathcal{C}^e = \frac{\phi\mathcal{E}}{\mathcal{W}}$ prob. met worker is u/e):

$$\begin{aligned} J^v(y, z, g) = & \mathcal{M}^v \mathcal{C}^u \int \alpha(\tilde{x}, y, z, g) (1 - \beta) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}} d\tilde{x} \\ & + \mathcal{M}^v \mathcal{C}^e \int \int \alpha^e(\tilde{x}, \tilde{y}, y) (1 - \beta) [S(\tilde{x}, y, z, g) - S(\tilde{x}, \tilde{y}, z, g)] \frac{g(\tilde{x}, \tilde{y})}{\mathcal{E}} d\tilde{x} d\tilde{y} \end{aligned}$$

- Free entry condition (entrants draw y from uniform density $\bar{h}^v(\tilde{y}) = 1$):

$$0 = -c + \int_0^1 J^v(\tilde{y}, z, g) \bar{h}^v(\tilde{y}) d\tilde{y}$$

- With homothetic $m(\mathcal{W}, \mathcal{V})$, invert to solve $\mathcal{V}$ as a function of (z, g) and $S(\cdot, \cdot, z, g)$.

On-the-job-search: KFE

- The KFE becomes:

$$\begin{aligned} dg_t(x, y) = & \underbrace{-\left(\delta(x, y, z) + \eta \alpha^b(x, y, z_t, g_t)\right) g_t(x, y) dt}_{\text{Outflow: exogenous \& endogenous separation}} \\ & - \underbrace{\mathcal{M}_t^e \int \alpha^e(x, y, \tilde{y}, z_t, g_t) \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} d\tilde{y} g_t(x, y) dt}_{\text{Outflow: workers poached to better matches}} \\ & + \underbrace{\mathcal{M}_t^u \alpha(x, y, z_t, g_t) \frac{g_t^v(y)}{\mathcal{V}_t} g_t^u(x) dt}_{\text{Inflow: unemployed workers hired}} \\ & + \underbrace{\mathcal{M}_t^e \int \alpha^e(x, \tilde{y}, y, z_t, g_t) \frac{g_t^v(y)}{\mathcal{V}_t} g_t(x, \tilde{y}) d\tilde{y} dt}_{\text{Inflow: workers poached from other matches}} =: \mu^g(x, y, z_t, g_t) dt \end{aligned}$$

Q4. Block recursive model impact on IRF to negative TFP shock?

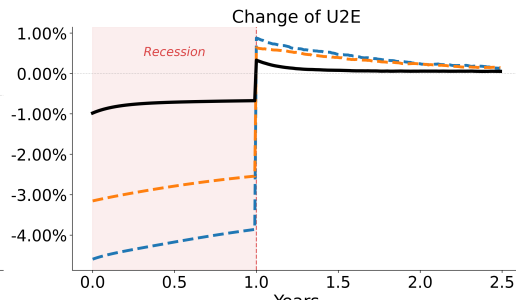
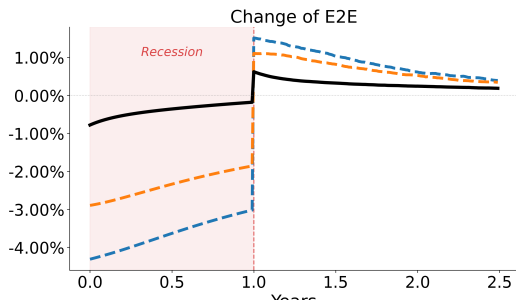
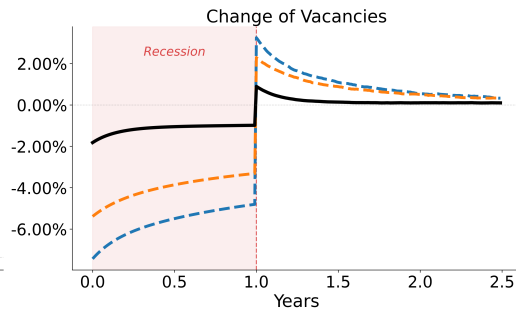
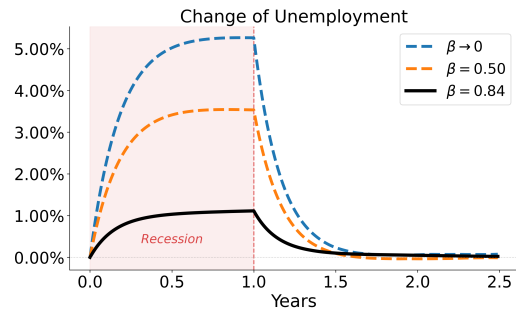


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On-The-Job Search Model with Idiosyncratic Shocks

- Introduce match specific (idiosyncratic) productivity shock ϵ (Mortensen and Pissarides, 1994) that follows a continuous time Markov chain with intensities $\lambda(\epsilon, \check{\epsilon})$.
- Production output from matches $F(x, y, \epsilon, z) = A_z A_\epsilon f(x, y)$; new matches draw initial $\epsilon \sim \pi(\epsilon)$.
- **Endogenous separation**: agents immediately break up the match when a new draw of ϵ makes the surplus negative ($\alpha^b(x, y, \epsilon, z, g) = \mathbb{1}\{S(x, y, \epsilon, z, g) < 0\}$).

On-The-Job Search Model with Idiosyncratic Shocks: Master Equation

$$\begin{aligned}\rho S(x, y, \epsilon, z, g) &= \rho(P(x, y, \epsilon, z, g) - V^u(x, z, g)), \quad V^v = 0 \text{ (short-lived vacancies)} \\ &= F(x, y, \epsilon, z) - \delta(x, y, z)S(x, y, \epsilon, z, g) - b \\ &+ \mathcal{M}^e \sum_{\tilde{\epsilon}} \pi(\tilde{\epsilon}) \int \alpha^e(x, y, \epsilon, \tilde{y}, \tilde{\epsilon}, z, g) \beta(S(x, \tilde{y}, \tilde{\epsilon}, z, g) - S(x, y, \epsilon, z, g)) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\ &- \mathcal{M}^u \sum_{\tilde{\epsilon}} \pi(\tilde{\epsilon}) \int \alpha(x, \tilde{y}, \tilde{\epsilon}, z, g) \beta S(x, \tilde{y}, \tilde{\epsilon}, z, g) \frac{g^v(\tilde{y})}{\mathcal{V}} d\tilde{y} \\ &+ \sum_{\check{\epsilon} \neq \epsilon} \lambda(\epsilon, \check{\epsilon}) ((1 - \alpha^b(x, y, \check{\epsilon}, z, g))(S(x, y, \check{\epsilon}, z, g) - S(x, y, \epsilon, z, g)) - \alpha^b(x, y, \check{\epsilon}, z, g) S(x, y, \epsilon, z, g)) \\ &+ \sum_{\check{z} \neq z} \lambda(z, \check{z}) (S(x, y, \epsilon, \check{z}, g) - S(x, y, \epsilon, z, g)) + \langle D_g S, \mu^g \rangle\end{aligned}$$

where $\alpha^e(x, y, \epsilon, \tilde{y}, \tilde{\epsilon}, z, g) = \mathbb{1}\{S(x, \tilde{y}, \tilde{\epsilon}, z, g) \geq S(x, y, \epsilon, z, g) \ \& \ S(x, \tilde{y}, \tilde{\epsilon}, z, g) \geq 0\}$.

On-The-Job Search Model with Idiosyncratic Shocks: KFE

$$\begin{aligned}
 dg_t(x, y, \epsilon) = & -\delta(x, y, z) g_t(x, y, \epsilon) dt \\
 & - \underbrace{\underbrace{\mathcal{M}_t^e}_{\text{Rate. e-worker meets}} \sum_{\tilde{\epsilon}} \pi(\tilde{\epsilon}) \int \underbrace{\alpha_t^e(x, y, \epsilon, \tilde{y}, \tilde{\epsilon})}_{\text{Prob. accept}} \underbrace{\frac{g_t^v(\tilde{y})}{\mathcal{V}_t}}_{\text{Prob. meet } \tilde{y}} d\tilde{y} \underbrace{g_t(x, y, \epsilon)}_{\text{Mass at } (x, y, \epsilon)}}_{\text{Mass accepting poaching offer}} dt \\
 & + \underbrace{\mathcal{M}_t^u}_{\text{Rate. u-worker meets}} \pi(\epsilon) \alpha_t(x, y, \epsilon) \frac{g_t^v(y)}{\mathcal{V}_t} g_t^u(x) dt \\
 & + \mathcal{M}_t^e \pi(\epsilon) \sum_{\tilde{\epsilon}} \int \alpha_t^e(x, \tilde{y}, \tilde{\epsilon}, y, \epsilon) \frac{g_t^v(y)}{\mathcal{V}_t} g_t(x, \tilde{y}, \tilde{\epsilon}) d\tilde{y} dt \\
 & - \sum_{\check{\epsilon} \neq \epsilon} \lambda(\epsilon, \check{\epsilon}) g_t(x, y, \epsilon) dt + \sum_{\check{\epsilon} \neq \epsilon} \lambda(\check{\epsilon}, \epsilon) (1 - \alpha_t^b(x, y, \epsilon)) g_t(x, y, \check{\epsilon}) dt
 \end{aligned}$$

(Last line: outflow when ϵ switches away; inflow from $(x, y, \check{\epsilon})$ when $\check{\epsilon}$ switches to ϵ , net of endogenous breakups.)

On-The-Job Search Model with Idiosyncratic Shocks: Results

- Recession experiment ($z = z_L$ for one year): cumulative net outflows from matches rise more, and recover more slowly, at the low idiosyncratic state ϵ_L (left)
... because low- ϵ matches carry lower surplus and sit near the breakup threshold $S = 0$.
- Endogenous separation flow $Q(x, y, z, g_{\text{erg}})$: recessions raise endogenous separations relatively more for high- (x, y) matches (right).

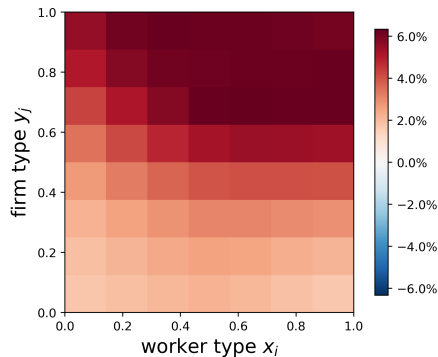
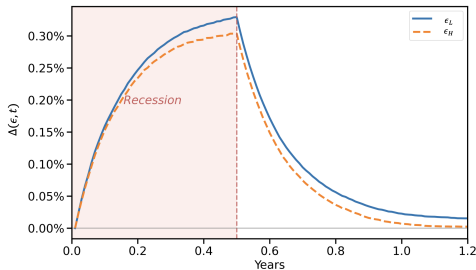


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Environment: Setting, Bonds, and Households

- Continuous time, infinite horizon environment.
- There are many bonds, $k \in \{1, \dots, K\}$, in positive net supply s_k :
 - Every bond pays the same dividend $\delta > 0$.
 - Bond k matures at rate $1/\tau_k$ (average maturity τ_k).
- Populated by a unit-mass continuum of infinitely-lived and risk-neutral investors:
 - An investor can hold either zero or one share of at most one type of asset.
 - Investor type $j \in \{1, \dots, J\}$ gets flow utility $\delta - \psi(j, k)$ from holding bond k .
 - Agents **switch type** from i to j at rate $\lambda_{i,j}$.
- Aggregate (default) state $z \in \{z_1, \dots, z_n\}$, switches at rate $\zeta_{z,z'}$.
At state z , asset k pays a fraction $\phi(k, z)$ of the coupon and the principal.

Distribution and Bargaining

- An investor's state is made up of her holding cost $j \in \{1, \dots, J\}$ and her ownership status, for each asset type $k \in \{1, \dots, K\}$ (owner o or non-owner n). Hence the set of investor idiosyncratic states is:

$$A = \{1n, 2n, \dots, Jn, 1o1, \dots, 1oK, 2o1, \dots, 2oK, Jo1, \dots, JoK\} \quad (3)$$

- The rate of contact between investors with states a and b is:

$$\mathcal{M}_{a,b} = \kappa_{a,b} g_a g_b \quad (4)$$

- Agents a, b **trade with each other**, engage in generalized Nash bargaining with bargaining power $\beta_{a,b}$.

Value Functions for Non-Owners and Owners [back](#)

- Value function for non-owner with type i satisfies:

$$\begin{aligned}\rho_i V(in, g, z) = & \sum_a \kappa_{in,a} \alpha(in, a, g, z) \beta_{in,a} S(in, a, z, g) \\ & + \sum_k \xi_{i,k} (V(iok, g, z) - V(in, g, z)) + \sum_{j \neq i} \lambda_{i,j} (V(jn, g, z) - V(in, g, z)) \\ & + \sum_{z'} \zeta_{z,z'} (V(in, g, z') - V(in, g, z)) + \sum_{a \in A} \partial_{g_a} V(in, g, z) \mu^g(a, z)\end{aligned}$$

- Value function for an investor of type i holding asset k satisfies:

$$\begin{aligned}\rho_i V(iok, g, z) = & \delta \phi(k, z) - \psi(i, k) + \frac{1}{\tau_k} (V(in, g, z) + \pi(k, z) - V(iok, g, z)) \\ & + \sum_a \kappa_{iok,a} \alpha(iok, a, g, z) g_a \beta_{iok,a} S(iok, a, g, z) + \sum_{j \neq i} \lambda_{i,j} (V(jok, g, z) - V(iok, g, z)) \\ & + \sum_{z'} \zeta_{z,z'} (V(iok, g, z') - V(iok, g, z)) + \sum_{a \in A} \partial_{g_a} V(iok, g, z) \mu^g(a, z).\end{aligned}$$

$\alpha(in, jok, g, z)$: indicator for whether the trade is accepted upon matching.

Endogenous Yield Curve and Impulse Responses

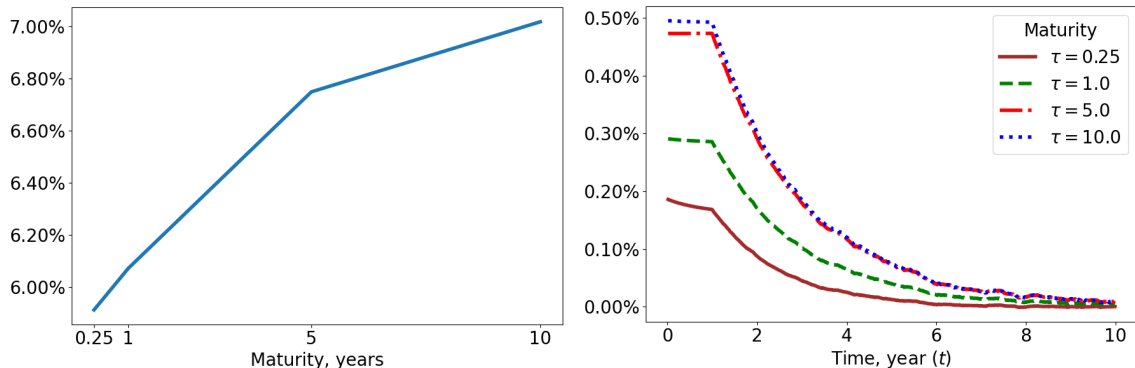
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Figure: Yield curve at ergodic steady state and impulse responses. Right figure: proportional bond yield change compared to the ergodic yield at each maturity following a one-year recession.

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Non-transferable Utility (NTU)

- Main change compared to the benchmark model: for a match (x, y) , agent x receives payoff $zf(x, y)$ and agent y receives $zf(y, x)$. No surplus to be split via bargaining.
- Computational implication:
 1. We no longer solve a single master equation for the match surplus.
 2. Instead, we directly approximate the value functions $V^u(x, z, g)$, $V^e(x, y, z, g)$, $V^v(y, z, g)$, and $V^p(x, y, z, g)$ using four neural networks, trained jointly to minimize a weighted sum of residuals from the four master equations.

Non-transferable Utility (NTU, Ct'd)

- Mean squared loss as a function of type in the master equations of V^u (left) and V^e (right)

